Data Structures

Queues and stacks

Understand

A queue is a collection that removes items in first-in, first-out order. Enqueue adds an item to the back of a queue. Dequeue removes the item at the front. A stack removes the most recently added item first.

Worked example

Queue trace: enqueue A, enqueue B, enqueue C. The queue is [A, B, C], with A at the front. Dequeue returns A and leaves [B, C]. A stack holding A, B, C with C on top would pop C.

Try it, then check.

Practice question

Enqueue red, blue and green, then dequeue twice. What remains?

Answer and reasoning

Green remains. Red leaves first, followed by blue.

Reflect

Explain the steps in your own words. Create a review card and return to page 1 when you need the original explanation.